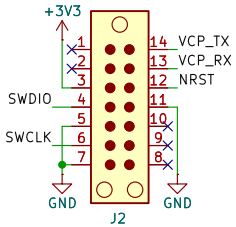
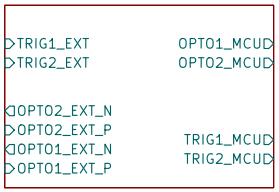
SWD Programming & UART Debug Interface

SWD flashing and VCP serial console connector
compatible with STDC14 / STLINK-V3

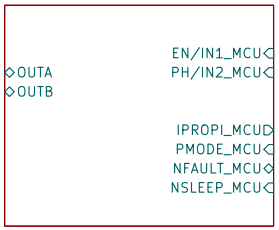


InterfaceSheet

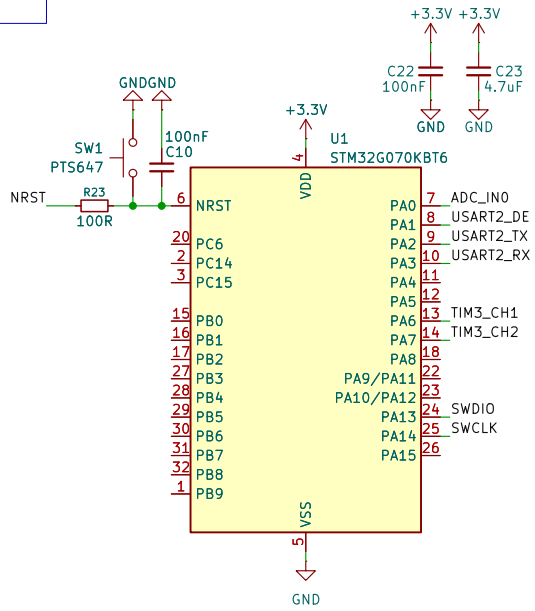


Fichier: Interface.kicad_sch

DriverMotorSheet



Fichier: DriverMotor.kicad_sch



PowerSheet



Fichier: Power.kicad_sch

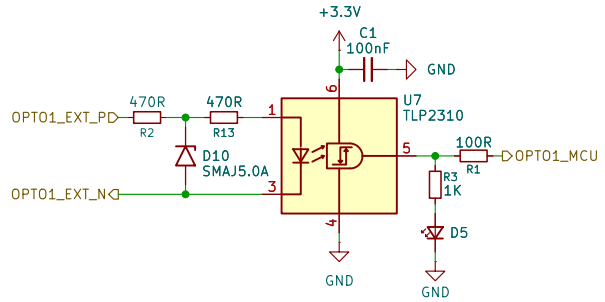
BerryRocket

Sheet: /			
File: KinetiCoreDC.kicad_sch			
Title: KinetiCoreDC – Main			
Size: A4	Date: 2026-10-09	Author : Paul Miailhe	**Rev: V°1**
KiCad E.D.A. 10.0.2			Id: 1/4

Text

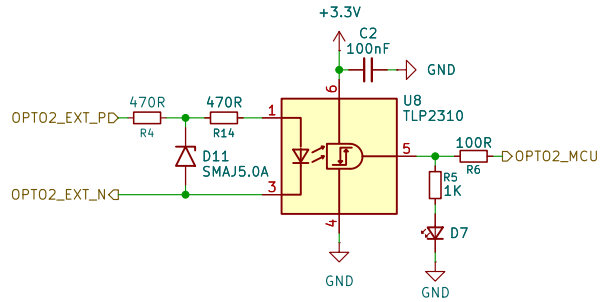
Optoisolated PWM Input N°1

Optoisolated input for secure PWM/GPIO control, protecting the logic circuit from DC motor noise.



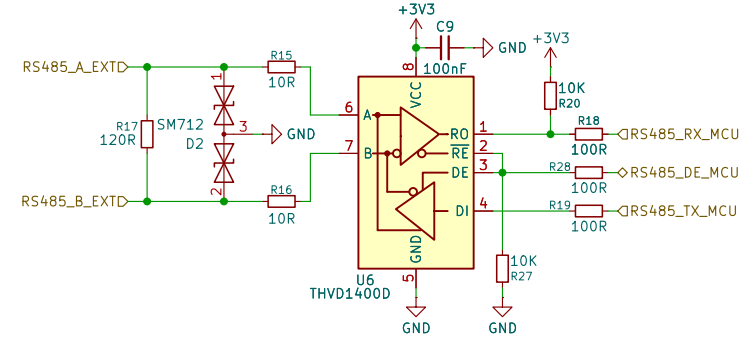
Optoisolated PWM Input N°2

Optoisolated input for secure PWM/GPIO control, protecting the logic circuit from DC motor noise.



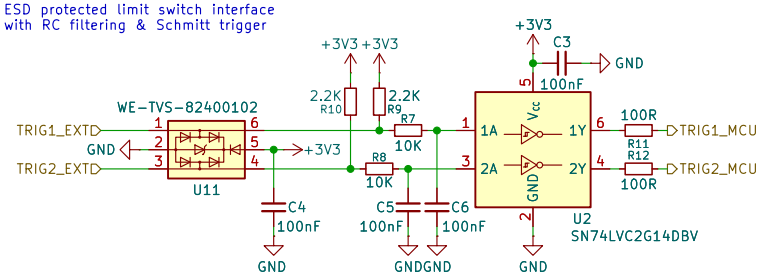
Protected Half-Duplex RS-485

Protected 3.3V half-duplex transceiver stage (up to 500 kbps)



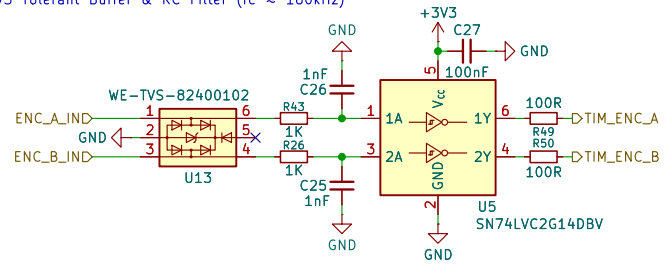
Limit Switch Input Conditioning

ESD protected limit switch interface with RC filtering & Schmitt trigger



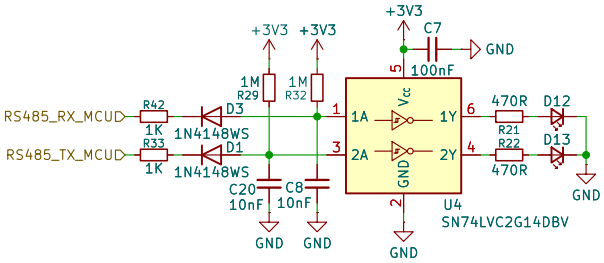
Encoder interface & ESD protection

Incremental Quadrature (A/B), 5V/3V3 Tolerant Buffer & RC Filter (fc ≈ 160kHz)



UART activity LED indicators (pulse stretcher)

Asymmetrical RC Monostable (10 ms Hold Time) & Schmitt-Trigger Buffer



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Sheet: /InterfaceSheet/
File: Interface.kicad_sch

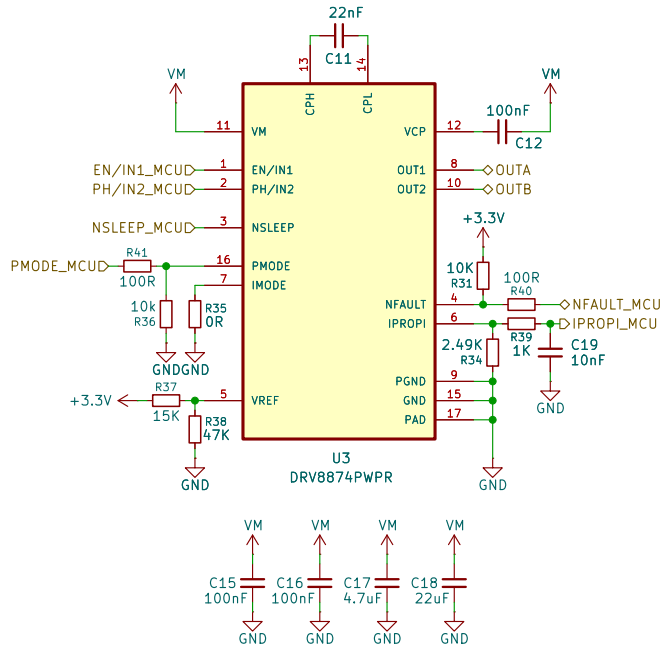
Title: KinetiCoreDC – Interface

Size: A4	Date: 2026-10-09	Author : Paul Miailhe	**Rev: V°1**
KiCad E.D.A. 10.0.2			Id: 2/4

Text

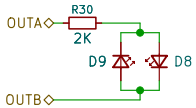
H-Bridge Motor Driver

Integrated DC motor driver with internal current sensing and regulation



Output Status Indicator

Bi-directional LED polarity and activity indicator across H-bridge outputs



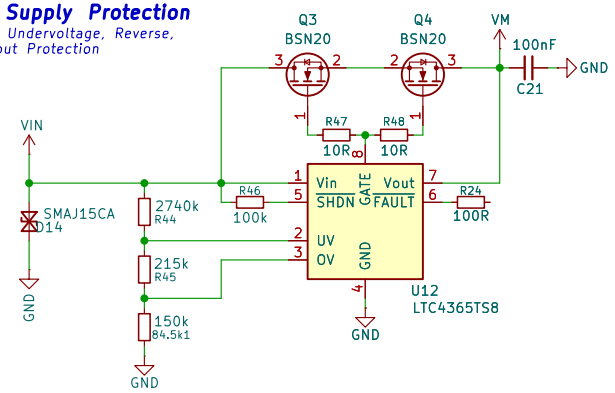
BerryRocket

Sheet: /DriverMotorSheet/			
File: DriverMotor.kicad_sch			
Title: KinetiCoreDC – DriverMotorDC			
Size: A4	Date: 2026-10-09	Author : Paul Miailhe	**Rev: V°1**
KiCad E.D.A. 10.0.2			Id: 3/4

Text

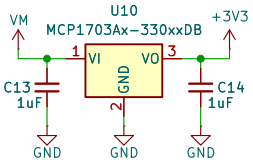
External Supply Protection

Overvoltage, Undervoltage, Reverse, and ESD Input Protection



Supply LDO Regulator

Low-quiescent-current 3.3V linear regulator for digital control logic



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Sheet: /PowerSheet/ File: Power.kicad_sch			
Title:			
Size: A4	Date:	Author :	**Rev:**
KiCad E.D.A. 10.0.2			Id: 4/4

Text